Regression (variance of ℓ_2 loss)

Var of ℓ_2 loss (relative to $t=0$)

2.3×

1.8×

Regression (RMSE of estimated ℓ_2 loss)

RMSE of estimated ℓ_2 loss (relative to $t=0$)

Classification (variance of log-loss)

Var of log-loss (relative to $t=0$)

3.5×

3.3×

Decision epoch t

Classification (RMSE of estimated log-loss)

RMSE of estimated log-loss (relative to $t=0$)

Decision epoch t

AUTODIFF 1-lookahead

REINFORCE (with baseline) 1-lookahead

REINFORCE (without baseline) 1-lookahead

Random sampling

Uncertainty sampling - static

Uncertainty sampling - sequential